



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to R.V.R. PETROLEUM AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

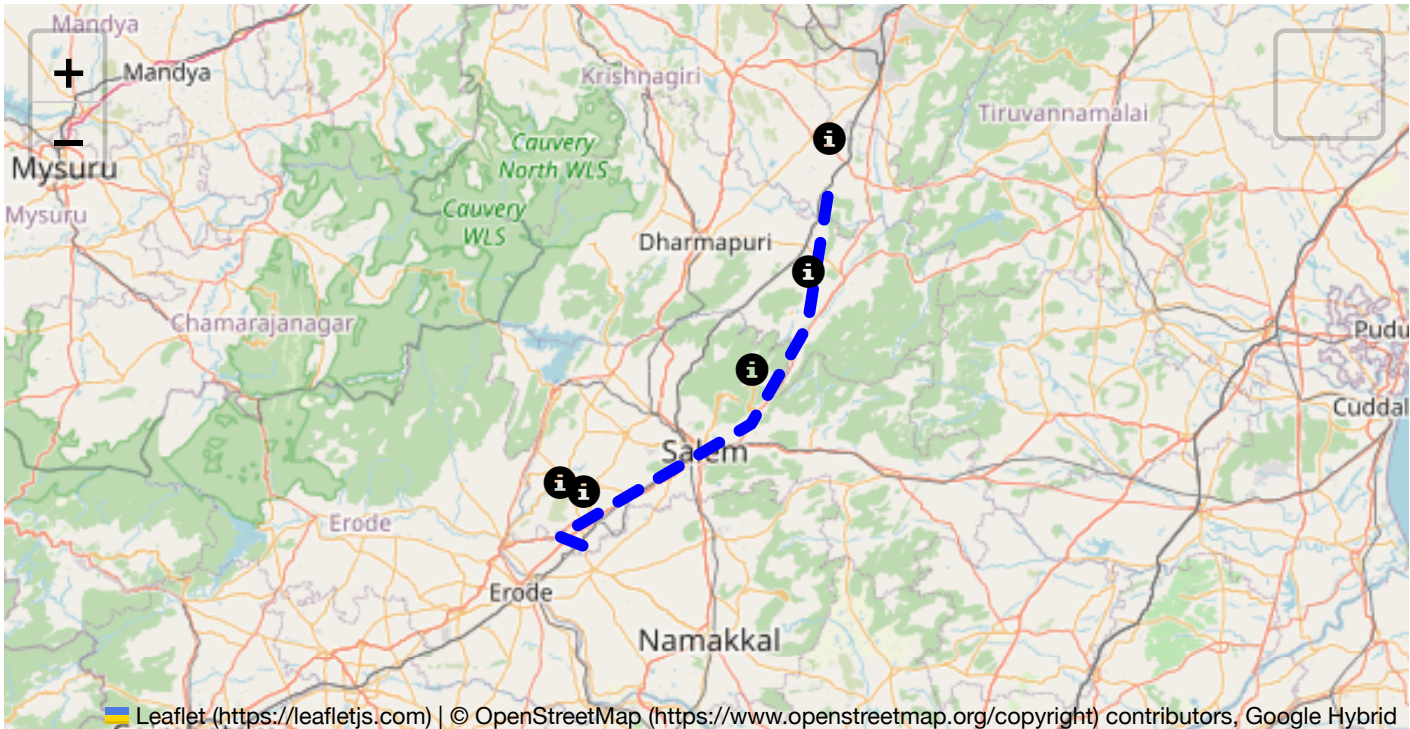
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

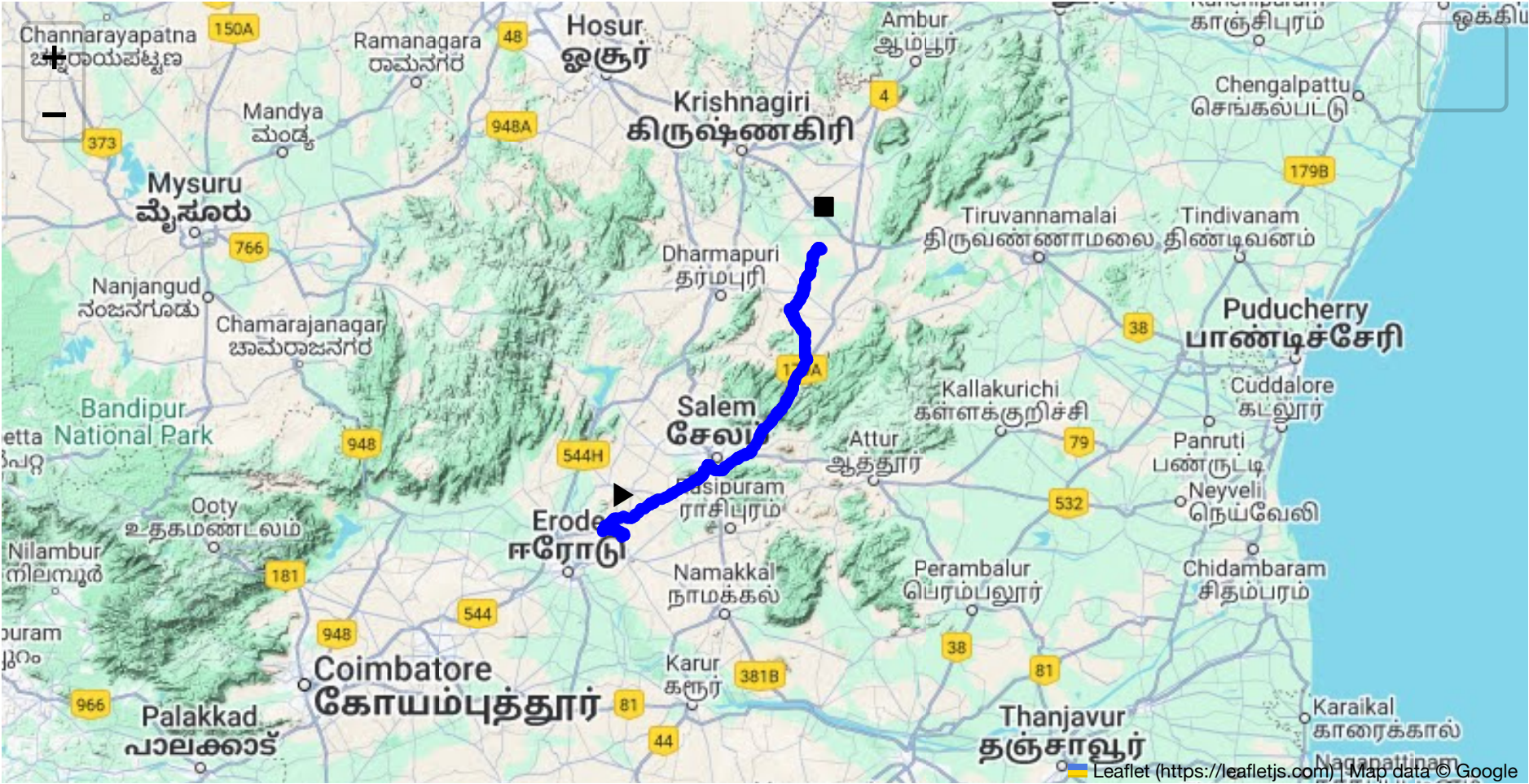
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 143.93 km
Estimated Duration: 3.5 hours
Adjusted Duration (Heavy Vehicle): 4.3 hours

Start: (11.4381, 77.8734)
End: (12.248695, 78.446663)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Kallavi via the specified waypoints covers approximately 144 kilometers and typically takes about 3 hours and 46 minutes for heavy vehicles carrying hazardous materials. This route traverses primarily through state highways and major roads, involving both urban and rural landscapes.

Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with hot summers, a monsoon season typically between October and December, and mild winters. During the monsoon, heavy rainfall can cause flooding, particularly in rural areas and low-lying regions, leading to road blockages and slippery conditions. It's prudent to watch for weather reports before traveling, especially during the monsoon season.

Analysis of Traffic Patterns

Traffic in this region can vary significantly. Urban areas like Sangagiri may experience congestion during peak hours (8-10 AM and 5-7 PM). Traffic diminishes as the route moves into rural areas. Bhavani Main Road can be busy due to local commutes and market days, typically Wednesday and Saturday, where congestion might increase.

Road Quality and Infrastructure

The road quality varies along this route. Major highways and state roads tend to be well-maintained, but care should be taken in rural areas where roads could be narrower and less well-maintained, with potential potholes. Additionally, watch for ongoing roadwork and deviations which are common in rural zones.

Alternative Routes for Emergencies

In case of emergencies, consider alternative routes based on the starting and endpoint. For example, bypassing rural areas by sticking close to larger highways may be advisable, though this could increase travel time. Utilize NH-44 for a more direct route if possible and safe.

Local Regulations on Hazardous Material Transport

Tamil Nadu has stringent regulations concerning the transport of hazardous materials. It is essential to have the necessary documentation, including permits and safety equipment. Driving through densely populated areas might attract more stringent vehicle checks.

Historical Incidents

There have been incidents involving heavy vehicles or hazardous materials, mostly related to poor road conditions or vehicle malfunction. Fatalities tend to occur mostly due to negligence at intersections or unprotected roadsides.

Environmental Considerations and Sensitive Areas

Environmental considerations include nearby agricultural land where contamination could be a concern. The region may have sacred sites or forested areas where spillages could have significant impacts. It is crucial to operate carefully in such regions.

Communication Coverage

Communication networks in urban and semi-urban areas are generally reliable. However, there may be dead zones, particularly in more remote sections between village areas. Drivers should be equipped with reliable communication tools, such as satellite phones for emergencies.

Estimated Emergency Response Times

The emergency response times can vary, with urban areas potentially seeing quicker response times (20-30 minutes) compared to remote areas where it might take 45 minutes to an hour. It is advisable to have first aid and emergency equipment on board.

Overall Summary of Risk Assessment

The route is moderately safe for truck transportation under routine conditions, but it requires caution in terms of weather, road quality, and rural traffic. Compliance with local regulations is essential to minimize risks. Monitoring weather conditions, having detailed maps, and ensuring reliable communication equipment can mitigate most potential hazards. Training drivers on local traffic norms and emergency procedures is also recommended to enhance safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45353, 77.85697	15 KM/Hr
5	Turn	Medium	11.45781, 77.85203	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	Medium	11.45546, 77.81800	30 KM/Hr
8	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
9	Turn	Medium	11.53578, 77.95802	30 KM/Hr
10	Turn	Medium	11.62802, 78.11548	30 KM/Hr
11	Turn	Medium	11.63087, 78.11738	30 KM/Hr
12	Turn	High	11.64144, 78.11803	15 KM/Hr
13	Turn	High	11.64220, 78.11887	15 KM/Hr
14	Turn	Medium	11.64210, 78.12083	30 KM/Hr
15	Turn	High	11.64152, 78.12164	15 KM/Hr
16	Turn	High	11.63917, 78.12086	15 KM/Hr
17	Turn	Medium	11.62860, 78.13285	30 KM/Hr
18	Turn	Medium	11.62901, 78.16571	30 KM/Hr
19	Turn	Medium	11.65803, 78.20452	30 KM/Hr
20	Turn	Medium	11.66354, 78.20645	30 KM/Hr
21	Blind Spot	Blind Spot	11.67120, 78.23136	10 KM/Hr
22	Turn	High	11.67171, 78.23139	15 KM/Hr
23	Turn	Medium	11.67462, 78.23933	30 KM/Hr
24	Turn	Medium	11.81243, 78.34087	30 KM/Hr
25	Turn	High	11.93914, 78.40426	15 KM/Hr
26	Turn	Medium	11.93979, 78.40377	30 KM/Hr
27	Turn	High	11.94043, 78.40189	15 KM/Hr
28	Turn	Medium	11.95570, 78.39975	30 KM/Hr
29	Turn	Medium	11.96453, 78.39648	30 KM/Hr
30	Turn	Medium	11.98782, 78.39989	30 KM/Hr
31	Turn	Medium	11.98885, 78.39938	30 KM/Hr
32	Turn	Medium	12.00565, 78.39738	30 KM/Hr
33	Turn	High	12.01030, 78.39704	15 KM/Hr
34	Turn	Medium	12.01072, 78.39781	30 KM/Hr
35	Turn	Medium	12.01286, 78.39888	30 KM/Hr
36	Turn	High	12.01431, 78.39858	15 KM/Hr
37	Turn	Medium	12.01550, 78.39613	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
38	Turn	Medium	12.06743, 78.36296	30 KM/Hr
39	Turn	Medium	12.07646, 78.35573	30 KM/Hr
40	Turn	High	12.07948, 78.35556	15 KM/Hr
41	Turn	Medium	12.08013, 78.35619	30 KM/Hr
42	Turn	Medium	12.09483, 78.37653	30 KM/Hr
43	Turn	Medium	12.10614, 78.38344	30 KM/Hr
44	Turn	High	12.10659, 78.38344	15 KM/Hr
45	Turn	High	12.10716, 78.38254	15 KM/Hr
46	Blind Spot	Blind Spot	12.12746, 78.39495	10 KM/Hr
47	Turn	High	12.13097, 78.39172	15 KM/Hr
48	Turn	Medium	12.14055, 78.39569	30 KM/Hr
49	Turn	Medium	12.14367, 78.39777	30 KM/Hr
50	Turn	Medium	12.19124, 78.41574	30 KM/Hr
51	Turn	High	12.19265, 78.41738	15 KM/Hr
52	Turn	Medium	12.20519, 78.41598	30 KM/Hr
53	Turn	Medium	12.22315, 78.41862	30 KM/Hr
54	Turn	Medium	12.22727, 78.41966	30 KM/Hr
55	Turn	High	12.22849, 78.41915	15 KM/Hr
56	Turn	High	12.23433, 78.42153	15 KM/Hr
57	Turn	Medium	12.23530, 78.42405	30 KM/Hr
58	Turn	High	12.25091, 78.43517	15 KM/Hr
59	Turn	High	12.25080, 78.43606	15 KM/Hr
60	Turn	Medium	12.25130, 78.43645	30 KM/Hr
61	Turn	High	12.25273, 78.43670	15 KM/Hr
62	Turn	High	12.25292, 78.43718	15 KM/Hr
63	Turn	High	12.25098, 78.43840	15 KM/Hr
64	Turn	High	12.24940, 78.43766	15 KM/Hr
65	Turn	Medium	12.24874, 78.43856	30 KM/Hr
66	Turn	High	12.24861, 78.43926	15 KM/Hr
67	Turn	High	12.25146, 78.44069	15 KM/Hr
68	Turn	Medium	12.25073, 78.44200	30 KM/Hr
69	Turn	High	12.24833, 78.44372	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
4	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
6	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
11	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
13	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
14	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
15	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
16	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
17	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
20	hospital	Dr.Thiru Neuro Multi Speciality	11.6623376, 78.203353	30 km/h	Medium
22	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
23	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
5	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
9	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
12	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
18	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
19	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium
21	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63303, 78.12713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53578, 77.95802



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62802, 78.11548



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63087, 78.11738



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64144, 78.11803



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64220, 78.11887



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64210, 78.12083



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64152, 78.12164



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63917, 78.12086



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62860, 78.13285



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62901, 78.16571



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65803, 78.20452



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66354, 78.20645



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.67120, 78.23136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67171, 78.23139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67462, 78.23933



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81243, 78.34087



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93914, 78.40426



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93979, 78.40377



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.94043, 78.40189



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95570, 78.39975



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96453, 78.39648



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98782, 78.39989



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98885, 78.39938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.00565, 78.39738



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.01030, 78.39704



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01072, 78.39781



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01286, 78.39888



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.01431, 78.39858



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01550, 78.39613



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.06743, 78.36296



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07646, 78.35573



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.07948, 78.35556



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.08013, 78.35619



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.09483, 78.37653



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.10614, 78.38344



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.10659, 78.38344



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.10716, 78.38254



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.12746, 78.39495



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13097, 78.39172



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14055, 78.39569



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14367, 78.39777



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19124, 78.41574



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.19265, 78.41738



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20519, 78.41598



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22315, 78.41862



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22727, 78.41966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.22849, 78.41915



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.23433, 78.42153



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.23530, 78.42405



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.25091, 78.43517



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.25080, 78.43606



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25130, 78.43645



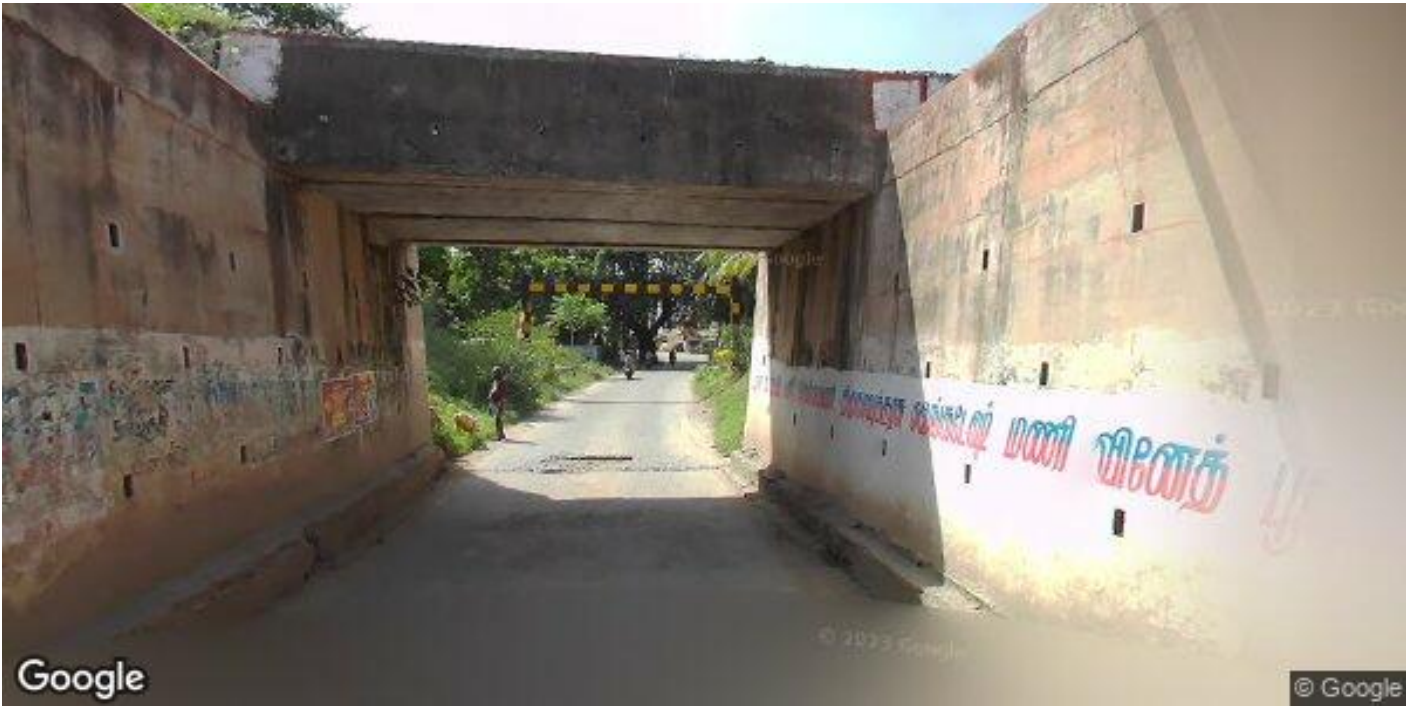
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.25273, 78.43670



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.25292, 78.43718



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.25098, 78.43840



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.24940, 78.43766



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24874, 78.43856



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.24861, 78.43926



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.25146, 78.44069

